Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 PHYSICS

9478/02

26 August 2026

2 hours

Paper 2 Structured Questions

You must answer on the question paper.
No additional materials are needed.

INSTRUCTIONS

- Answer all questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and index number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do not use an erasable pen. Do not use correction fluid or tape.
- Do not write on any bar codes.
- You may use an approved calculator.

- **INFORMATION**
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use	
1	/10
2	/8
3	/14
4	/5
5	/10
6	/8
7	/20
s.f.	/−1
Total	/75

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\mathrm{m\,F^{-1}})$
elementary charge	$e = 1.60 \times 10^{-19}\,\mathrm{C}$
Planck constant	$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\mathrm{kg}$
rest mass of electron	$m_\mathrm{e} = 9.11 \times 10^{-31}\,\mathrm{kg}$
rest mass of proton	$m_\mathrm{p} = 1.67 \times 10^{-27}\,\mathrm{kg}$
molar gas constant	$R = 8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Avogadro constant	$N_\mathrm{A} = 6.02 \times 10^{23}\,\mathrm{mol^{-1}}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
acceleration of free fall	$g = 9.81\,\mathrm{m\,s^{-2}}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\mathrm{K} = T/^\circ\mathrm{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

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displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

1 An object of mass 0.050 kg is projected from the top of a 50 m tall building at an angle of 60° to the horizontal with an initial speed of 20 m s^{-1} as shown in Fig. 1.1.

A is a point in the trajectory that is at the same height as the object initially and *d* is the horizontal distance the object lands from the building.

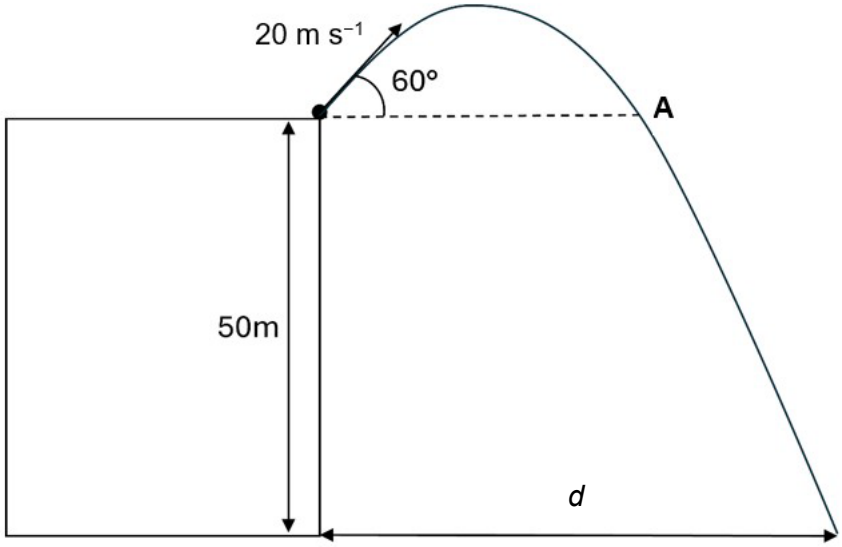


Fig. 1.1

(a) Assuming there is no resistance.

(i) Calculate the distance *d*.

d = ... m [3]

(ii) Without calculation, state the speed of the object at **A** and explain how you arrived at your answer.

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…… [2]

(b) Experimentally, it is found that the same object now reaches a maximum height of 60.5 m above the ground in its trajectory.

(i) Show quantitatively that the air resistance is not negligible in this case.

[2]

(ii) The speed of the object at its maximum height is 8.0 m s^{-1}.
Calculate the energy lost due to air resistance at maximum height.

energy lost = ... J [2]

(c) Sketch the path of the object in Fig. 1.1 when air resistance is not negligible. [1]

[Total: 10]

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2 In a collision between two bodies A and B, the force that A exerts on B varies with time in the way shown in Fig. 2.1.

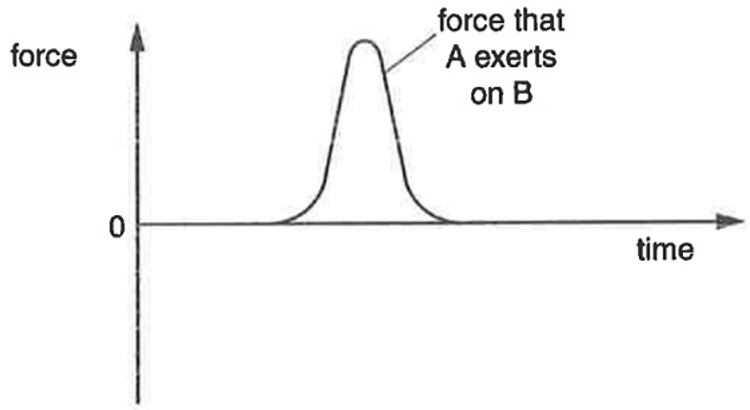


Fig. 2.1

(a) **(i)** Copy Fig. 2.1 and show in Fig. 2.2 a graph of the force that B exerts on A.



Fig. 2.2 [1]

(ii) Explain your answer to **(i)**.

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……..…… [1]

(b) In a collision, when a truck of mass 12 000 kg runs into the front of a car of mass 1200 kg, a constant force of 72 000 N acts for 0.25 s.

Calculate the change in velocity of

(i) the car,

change in velocity = ... m s^{-1} [2]

(ii) the truck,

change in velocity = ... m s^{-1} [2]

(c) State and explain how the force on the driver could be reduced by safety features in a typical car.

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……..…… [2]

[Total: 8]

3 An experiment is carried out, using a light intensity meter attached to a toy train, to measure the diffraction of laser light through a double slit, as shown in Fig. 3.1 below.

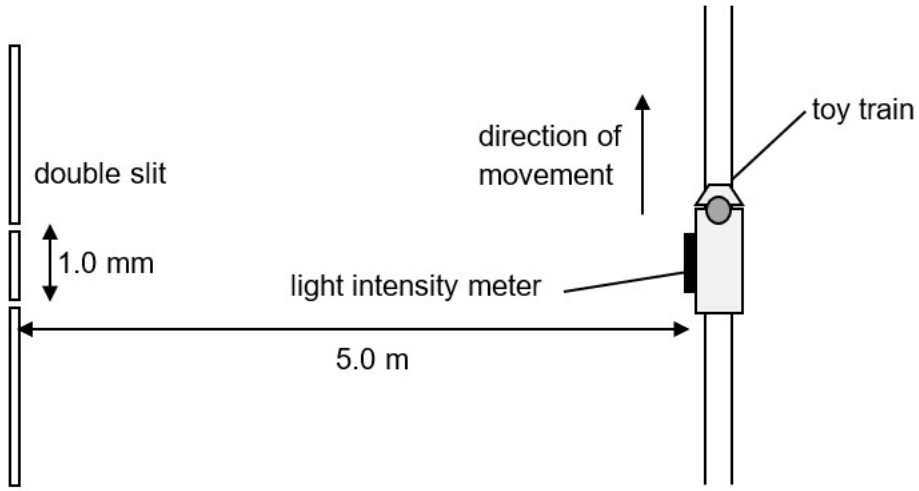


Fig. 3.1

Light from the laser is incident on the double slit. The light from the double slit is then detected by a light sensor attached to the train.

As the train moves along the track, the light intensity is recorded by the light intensity meter. The variation with time of the light intensity is shown in Fig. 3.2.

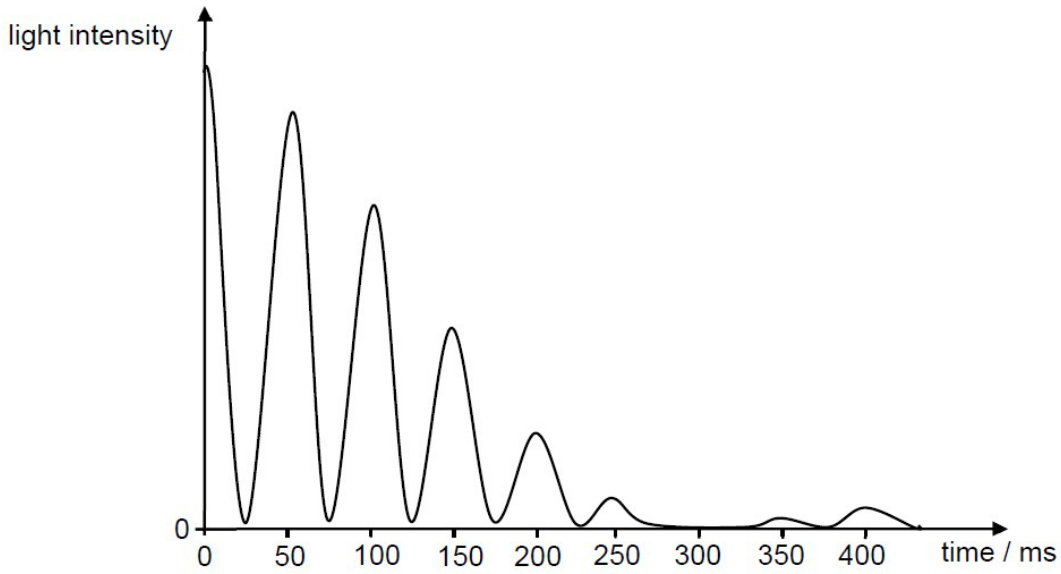


Fig. 3.2

(a) Explain, with reference to the paths travelled by the light emerging from each slit, why maxima and minima are observed.

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……..… [2]

(b) **(i)** The speed of the train is 6.5 cm s^{-1}.

Use Fig. 3.2 to show that the fringe separation of the two slit interference pattern is 3.3×10^{-3} m.

[1]

(ii) The separation of the slits is 1.0 mm and the track is 5.0 m from the slits.

Determine the wavelength of the laser light.

wavelength = ... m [2]

(c) A maximum is detected by the light intensity meter at every 50 ms, explain why a minimum is detected at 300 ms in Fig. 3.2.

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……..…… [1]

(d) Determine the width of one of the slits.

width = ... m [2]

(e) The experiment is changed so that the toy car is now used to investigate the formation of a stationary sound wave as shown in Fig. 3.3 below.

The light sensor of the train is replaced with a sound sensor attached to a data logger. A loudspeaker emitting sound waves of constant amplitude and frequency is placed such that it faces a reflecting barrier. The train track is adjusted so that the train is moving away from the loudspeaker towards the barrier.

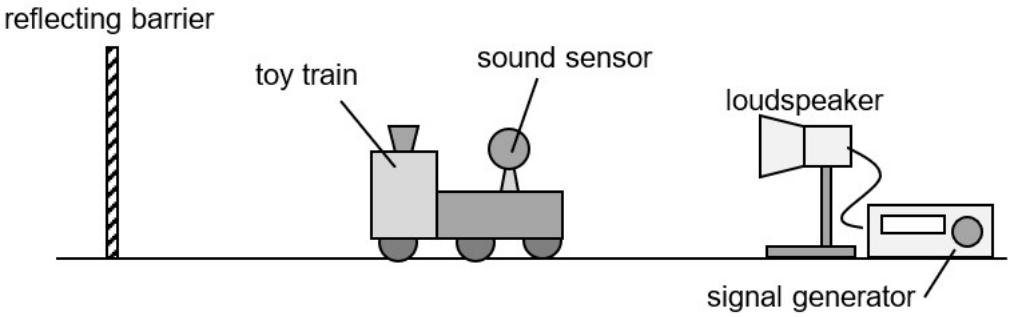


Fig. 3.3

The variation with distance of the sound intensity recorded by the sound sensor is shown in Fig. 3.4.

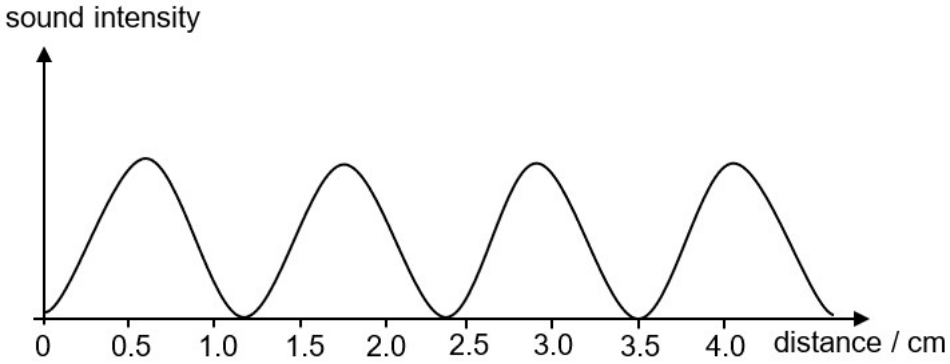


Fig. 3.4

(i) Explain how a stationary sound wave forms between the loudspeaker and reflecting barrier.

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…… [2]

(ii) Using Fig. 3.4, determine the wavelength of the stationary wave.

wavelength = ... m [2]

(iii) Given that the frequency of the sound is 14 kHz.
Determine the speed of the sound wave.

speed = ... m s^{-1} [2]

[Total: 14]

4 A technician is designing a charging circuit for a compact camera flash unit as shown in Fig. 4.1. The circuit consists of a DC battery V of 12 V, a resistor R = 15 kΩ and an uncharged capacitor C of 220 μF, connected in series with a switch.

At time t = 0, the switch is closed.

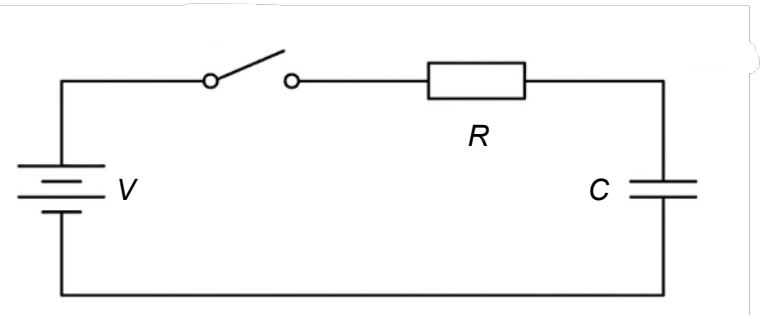


Fig. 4.1

(a) Calculate the time constant for the circuit.

time constant = ... s [1]

(b) Calculate the percentage of maximum charge in the capacitor 6.6 s after the switch is closed.

percentage = ... % [2]

(c) Hence or otherwise, determine the current flowing through the circuit exactly 6.6 s after the switch is closed.

current = ... mA [2]

[Total: 5]

5 A circuit is set up as shown in Fig. 5.1. The circuit contains a battery of e.m.f. 12 V, and a fixed resistor r of resistance 6 Ω and length L_r = 0.50 m. A movable metal rod M, of length L_M = 0.60 m and resistance 12 Ω, is placed on a conducting track at a distance d = 4.0 cm from r. The conducting track is assumed to be smooth and has zero resistance.

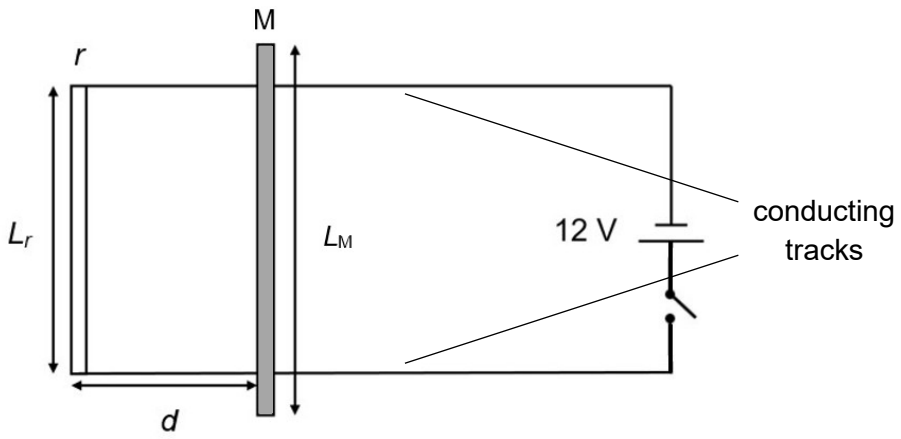


Fig. 5.1 (not to scale)

The switch is now closed.

(a) Show that the magnetic flux density B at the metal rod M due to the current in resistor r is 10 μT.

[1]

(b) Calculate the magnetic force experienced by M and state its direction.

magnetic force = ... N

direction: ..... [3]

(c) M in **(b)** is held stationary and a magnetic field of flux density 1.0 mT is now directed out of page throughout the circuit in Fig. 5.1. M is now released.

Describe and explain the initial motion of M.

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…… [2]

(d) A proton enters the magnetic field *B* at an angle to the field of less than 90°.

The proton follows the path shown in Fig. 5.2.

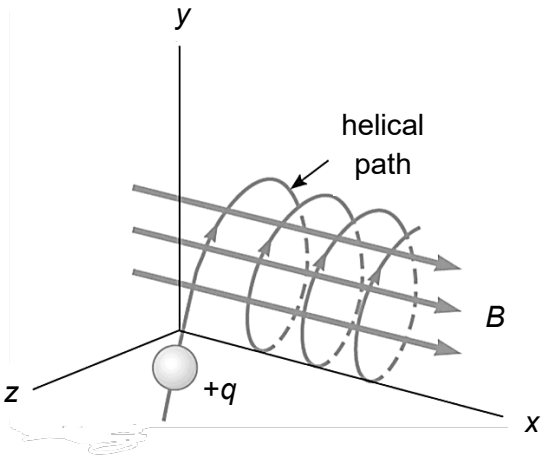


Fig. 5.2

(i) Explain why the proton follows this helical path.

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…… [2]

(ii) Fig. 5.3 shows a magnetic bottle, where a non-uniform magnetic field is produced between two current carrying coils. A proton is initially directed perpendicularly into the page with speed v at the position shown.

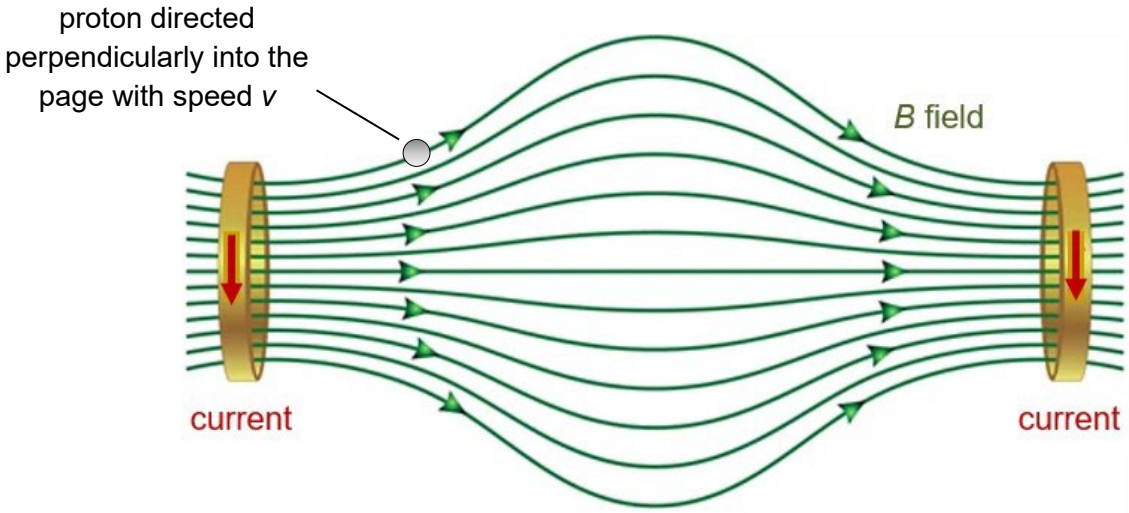


Fig. 5.3

Describe and explain the subsequent motion of the proton in the magnetic bottle, assuming the proton remains within the field region throughout.

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…… [2]

[Total: 10]

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6 (a) (i) Explain what is meant by the term *nuclear binding energy*.

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.....[2]

(ii) A nucleus of strontium-90 has a mass of 89.907 *u*. It contains 38 protons. The mass of a neutron is 1.00867 *u* and the mass of a proton is 1.00783 *u*.

Calculate the nuclear binding energy per nucleon of strontium-90.

binding energy per nucleon = ... J [2]

(b) The nuclear fission of uranium-235 can be described by

$$^{1}_{0}n + {}^{235}_{92}U \longrightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3{}^{1}_{0}n.$$

Table 6.1 shows the masses of the particles involved in this reaction.

Table 6.1

particle	mass / u
$^{1}_{0}n$	1.008665
$^{89}_{36}Kr$	88.917636
$^{144}_{56}Ba$	143.922953
$^{235}_{92}U$	235.043930

Determine the energy released by the fission of 15.0 kg of uranium-235.

energy = ... J [4]

[Total: 8]

7 Read the passage below and answer the questions that follow.

Traditional wind turbines are limited by their height; the friction of the Earth's surface slows wind speeds at low altitudes, and the massive concrete foundations required for towers over 150 metres tall are both expensive and environmentally invasive. To overcome these limitations, engineers have developed **Airborne Wind Energy Systems (AWES)**.

In early 2026, the S2000 Stratosphere Airborne Wind Energy System (SAWES), developed in China, became the world's first megawatt-class "flying power plant" to successfully feed electricity into a national grid. The system utilizes a helium-filled, ring-shaped aerostat (a "balloon") anchored to a ground station by a high-strength tether. A simplified schematic of the S2000 system is shown in Fig. 7.1. The aerostat remains aloft through upthrust, the upward force exerted by the surrounding air on the balloon when it is immersed in the air. This allows the turbines to access the "jet effect" of the stratosphere—a region where winds are faster, more stable, and carry significantly more kinetic energy than those at ground level.

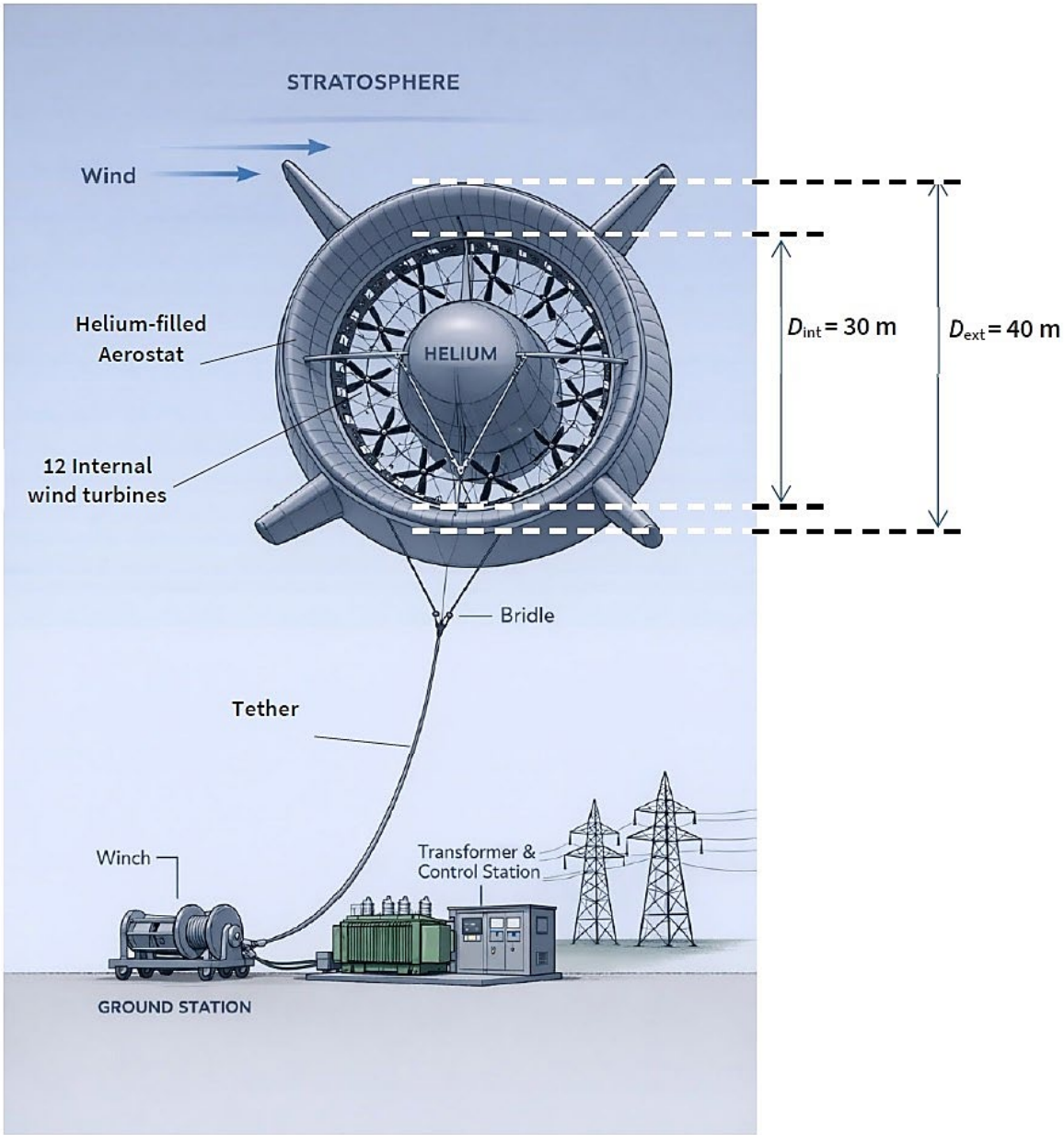


Fig. 7.1 Simplified Schematic of the S2000 System

The S2000 is linked to its ground-based winch station by a 2.5 km multipurpose conductive tether. This cable is a composite structure, featuring a high-tensile carbon-fibre core to withstand buoyancy and wind-drag forces, surrounding an aluminium-based conductive layer for power transmission. Onboard sensors relay real-time data through fibre-optic filaments within the tether, allowing a ground-based AI to adjust turbine speeds and aerostat pitch.

Table 7.1: S2000 SAWES System Specifications and Operating Parameters

Category	Parameter	Value (at h = 2000 m)
Atmospheric	Air density (ρ_{air})	1.007 kg m^{-3}
	Average Wind Speed (v)	15.0 m s^{-1}
Aerostat	External Diameter (D_{ext})	40 m
	Internal Diameter (D_{int})	30 m
	Volume of Helium (V_{He})	20,000 m^3
	Aerostat Structural Mass ($m_{structure}$)	5,000 kg
Generation	Total Turbine Swept Area ($A_{turbine}$)*	340 m^2
	Turbine Assembly Mass ($m_{turbine}$)*	2,400 kg
	Maximum Rated Power (P_{max})	3.0 MW
	Average Power Output from Initial Test	770 kW (over 30 min)
Tether	Total Length l_{tether}	2500 m
	Linear Mass	1.8 kg m^{-1}

*Note: Total of the 12 internal turbines.

Additional Constants and Parameters:

Aluminium resistivity: $\rho_{Al} = 2.8 \times 10^{-8}\ \Omega$ m

Helium density: $\rho_{He} =$ 0.18 kg m^{-3}

Power coefficient: $C_p = 0.40$ (maximum)

System efficiency: $\eta_{system} = 45\%$ (maximum)

Drag coefficient (ring aerostat): $C_d = 0.25$

Energy density of the stratospheric wind, which represents the kinetic energy per unit volume of moving air, helps engineers assess the energy resource available at different altitudes and locations. This, in turn, will help to determine the amount of power that can be harnessed from the wind through the turbine.

Real turbines cannot extract all available wind power. The power coefficient (C_p) represents the fraction of wind power actually converted to mechanical power. For example, a C_p of 0.40 means the turbine converts 40% of the wind's kinetic energy into mechanical rotation. System efficiency (η_{system}) accounts for subsequent losses during electrical conversion and transmission, reducing the final delivered power. This includes losses in the generator, power electronics, and the resistance of the 2.5 km tether. A system efficiency of 45% means only 45% of the mechanical power extracted by the turbines reaches the ground station as usable electricity.

(a) **(i)** Using data from Table 7.1, calculate the buoyant force (upthrust) $\rho_{air} V_{He} g$ acting on the helium-filled aerostat at its operating altitude.

buoyant force = ……. N [1]

(ii) Calculate the total weight of Helium-filled aerostat.

total weight = ……. N [2]

(iii) Using your answers from **(a)(i)** and **(a)(ii)**, determine the minimum tension in the tether.

minimum tension = ……. N [1]

(iv) Calculate the weight of the tether.

weight = ……. N [1]

(v) State whether the force exerted by the Winch on the tether will be greater, equal or smaller than your answer in **(a)(iii)**, explain your answer.

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……..

……..… [2]

(b) **(i)** When wind flows around the ring-shaped aerostat, it experiences a drag force given by $\frac{1}{2}\rho v^2 C_d A$, where A is the frontal area presented to the wind. Assuming the frontal area is a circle with external diameter D_{ext}, calculate the drag force acting on the aerostat in the stratospheric wind conditions.

drag force = ……. N [2]

(ii) Determine the magnitude of the change in tension in the tether, when drag force acts on the aerostat. Explain your answer.

magnitude of change in tension = ……. N [2]

(c) **(i)** By considering the energy density of the stratospheric wind, show that the power that can be harnessed from the wind can be expressed as $\frac{1}{2}\rho_{air}A_{turbine}v^3$, where ρ_{air}, $A_{turbine}$ and v are quantities expressed in Table 7.1.

[2]

(ii) Determine the total power harnessed from the wind through the turbine.

total power = ……. W [1]

(iii) Hence or otherwise, determine maximum electrical power delivered to ground.

power = ……. W [2]

(iv) The average power output from the Initial test from Table 7.1 is 770 kW. State whether this is higher or lower than your calculated value in **(c)(iii)** and suggest a reason for the difference.

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…… [2]

(d) Dimensions of the tether must be considered in order to carry a current of up to 300 A and heat losses through the tether only account for 0.48% of the maximum rated power $\left(P_{\text{max}}\right)$ for efficient transmission of power.

Determine the cross-sectional area of the aluminium needed in the conductive layer of the tether that satisfies the above considerations.

cross-sectional area = …….…… m^2 [2]

[Total: 20]

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